

Combine the analysis of Gene Expression Omnibus datasets with the OmniSearch semantics-driven analysis to computationally discover microRNA-related biomarkers in glioblastoma

Huawei Jin^{1*}, Zhenhua Yu^{1*}, Xiaodan Wang², Weian Chen³, Shaolei Guo⁴, Mohan Vamsi Kasukurthi⁵, Shengyu Li⁵, Glen M. Borchert⁶, Jingshan Huang^{5†}

¹Department of Neurosurgery, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou, Guangdong, China

²School of Nursing, Sun Yat-sen University, 58 Zhongshan Road II, Guangzhou, Guangdong, China

³Department of Nuclear Medicine, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou, Guangdong, China

⁴Department of Neurosurgery, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou, Guangdong, China

⁵School of Computing, University of South Alabama, Mobile, Alabama 36688, U.S.A.

⁶College of Medicine, University of South Alabama, Mobile, Alabama 36688, U.S.A.

*These authors contributed equally to this work

†Corresponding author

Email addresses:

HJ: jinhw@mail.sysu.edu.cn

ZY: yuzhenhua08@126.com

XW: wangxdan@mail.sysu.edu.cn

WC: cwa909@163.com

SG: guoshaol@mail.sysu.edu.cn

MVK: mk1530@jagmail.southalabama.edu

SL: sl1721@jagmail.southalabama.edu

GMB: borchert@southalabama.edu

JH: huang@southalabama.edu

Abstract

Background

Glioblastoma multiforme is the most malignant and common primary brain tumor with rapid relapse. Biomarker discovery for early detection of glioblastoma is still a primary goal to improve its diagnosis, treatment, and prognosis. It is known that highly stable extracellular microRNAs circulate in the blood of both healthy individuals and patients, thus affording opportunities for real-time monitoring of tumor burden and therapeutic response in many tumors. However, such utility is rather limited in glioblastoma multiforme.

Methods

We first analyzed differentially expressed microRNAs out of three microRNA datasets, each consisting of multiple expression profiles obtained from the Gene Expression Omnibus repository. Next, we used Venn diagrams to initially filter out more likely regulating microRNAs. Finally, we utilized OmniSearch, a semantic search tool, to further filter out most promising microRNAs.

Results

750 differentially expressed microRNAs (250 from each of three microRNA datasets) were obtained, among which ten microRNAs were commonly identified differentially expressed microRNAs from all datasets. After semantic searches and analysis, three microRNAs were discovered to be closely related to glioblastoma multiforme.

Conclusions

We developed comprehensive methodologies, which include the analysis of Gene Expression Omnibus datasets as well as semantics-driven data analysis, to help better identify likely regulating microRNAs in glioblastoma multiforme. Our results concluded that hsa-miR-142-3p, hsa-let-7c, and hsa-miR-140-5p are potentially novel biomarkers in glioblastoma multiforme.

Background

According to the current World Health Organization (WHO) guidelines, gliomas are subcategorized by histopathologic evaluation into four tumor grades (Grades I through IV). Glioblastoma multiforme (GBM) is the most malignant (Grade IV) and most common primary brain tumor with high morbidity [1]. Due to the unique location, aggressive biological behavior, and diffuse infiltrative growth, there exist many critical challenges in the treatment of GBM [2][3]. Despite comprehensive therapy strategies that include aggressive surgical resection, chemotherapy, and radiation, median survival of GBM patients remains approximately 14 months, and lethality for most patients is less than two years [4].

Currently, histological examination of the tumor tissue is the standard procedure for definitive diagnosis of gliomas, while computer tomography (CT) and nuclear magnetic resonance imaging (MRI) are supplement procedures for disease staging. Unfortunately, both the histopathology and neuroimaging tests are not only expensive but also rather insensitive and may cause hemorrhage and neurologic damages. Thus, major interest has risen in developing biomarkers that allow for less expensive monitoring of the disease within shorter time intervals [5]. Furthermore, patients at high risk for surgery-associated mortality or having small tumors in eloquent areas of the brain could significantly benefit from the discovery of biomarkers so that they can avoid biopsy for the confirmation of GBM [6].

Recent studies [7][8] suggested that GBM cells secrete extracellular vesicles (EVs) containing genetic materials that mirror the intracellular tumor milieu. In addition, sampling of these vesicles derived from biofluids such as sera or cerebrospinal fluid (CSF) has been proposed [9] as a means of “liquid biopsy” that affords opportunities for real-time monitoring of therapeutic response and cancer burden (a.k.a. tumor load, meaning the number of cancer cells, the size of a tumor, or the amount of cancer in the body). Although early detection with circulating biomarkers is an established method in the diagnosis and treatment of many different cancer types, diagnostic tests for GBM are currently still quite limited [10].

To effectively obtain a set of biomarkers for early detection is still a primary goal to improve diagnosis and treatment of GBM. In particular, abnormal expression of microRNAs (miRNAs or miRs) was found to be closely associated with GBM disease progress [11-16], and a better understanding of miRNA regulation mechanisms may result in the identification of novel biomarkers in GBM. It is known that miRNAs perform their functions through respective target mRNAs. However, to computationally identify appropriate miRNAs, study specific miRNA functions, and investigate miRNA::mRNA regulation mechanisms are very complicated. Research efforts in this regard include, but are not limited to, (i) to query the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) repository [17] for microarray and other forms of high-throughput gene expression data; (ii) to search for biologically validated miRNA targets from the PubMed database [18]; (iii) to find computationally putative miRNA targets by inquiring on numerous prediction databases; and (iv) to obtain additional information for each miRNA target, whether validated or predicted, from many distributed data sources such as Gene Ontology (GO) [19] annotations, RNAcentral [20] annotations, and pathway analysis by Protein ANalysis THrough Evolutionary Relationships (PANTHER) Classification System [21]. Along this research line, there are two important observations. (1) First, we are dealing with large amounts of biological, biomedical, and clinical data. (2) Second, and possibly more importantly, all involved data sources have been developed

by different research labs/groups, thus having quite heterogeneous semantics. As a result, conventional knowledge acquisition and discovery approaches (i.e., those approaches that are not driven by data semantics) are not suitable to handle complex data semantics contained in such big data relevant to miRNA research. Therefore, it is necessary to incorporate emerging semantic technologies into conventional miRNA knowledge discovery to achieve more effective and more efficient miRNA knowledge acquisition.

In our previous research efforts, we have been developing OmniSearch [22-24], a data integration and analytical software tool that was specifically designed for miRNA research based upon domain ontologies and semantic technologies. We report in this paper our innovative methodologies, which combine the GEO data analysis with the OmniSearch semantic search and analysis in a seamless manner, to computationally identify appropriate serum miRNAs as effective biomarkers in the diagnosis of GBM.

Methods

Our methodologies include three steps.

- 1) Step 1: Identification and analysis of differentially expressed miRNAs (DEMs) in GBM out of miRNA expression profiles obtained from the GEO repository.
- 2) Step 2: Initial screening of more likely regulating miRNAs in GBM via the application of Venn diagrams.
- 3) Step 3: Further screening and filtering of regulating miRNAs in GBM through semantic searches and analysis in the OmniSearch software tool.

A. Step 1: DEM identification out of GEO datasets.

The GEO repository archives and freely disseminates microarray and other forms of high-throughput data generated by the scientific community. With a minimum information about a microarray experiment (MIAME)-compliant infrastructure, fully annotated raw and processed data are contained in the GEO database [17]. After obtaining miRNA expression microarray datasets of GBM from GEO, we used the GEO2R tool in the GEO repository to perform comparisons on original submitter-supplied processed data. GEO2R is an interactive Web tool that allows users to compare two or more groups of samples in GEO datasets (i.e., series) to identify genes and/or RNA molecules that are differentially expressed across experimental conditions. In the current study, when comparing different expression levels of miRNAs between the GBM and control groups, we adopted two thresholds following common practice in statistics, i.e., (i) adjusted P value (*adj.P.Val*) of less than 0.05 and (ii) absolute log fold change (*logFC*) of greater than 1.0, to identify DEMs out of original GEO series, and the resultant DEMs were presented in a table sorted by significance (descending order on *adj.P.Val*). In case the number of miRNAs was less than 250, we included top 250 miRNAs based on their adjusted P values.

Note that the number of identified DEMs from GEO datasets is usually, if not always, quite large, resulting in a rather challenging analysis process. In some cases, the analysis cannot even proceed. Therefore, it is necessary to filter out more promising DEMs for an effective investigation on miRNA function and regulation.

B. Step 2: Initial miRNA screening via Venn diagrams.

A Venn diagram is also referred to as a primary/set/logic diagram, which is able to show possible logical relations among a finite collection of different sets [25]. Note that one of the most prevalent logical relations is the intersection relation. Venn diagrams have been utilized in various research fields including biological and biomedical areas. In the present study, we adopted Venny 2.1 [26], an interactive tool for comparing sets with Venn diagrams, during the initial screening on DEMs retrieved from GEO. Specifically, we focused on the intersection among DEMs obtained from numerous GEO series; that is, we selected those DEMs commonly identified by all involved GEO series – the idea being that, with the support of more GEO series, such DEMs are more likely to actually prove meaningful ones. Also note that DEMs identified from GEO include both over-expressed and under-expressed miRNAs.

After this initial screening through Venn diagrams, the number of computationally putative regulating miRNAs was significantly reduced. However, there was room for further improvement. Recall that miRNAs perform their functions through respective targets, and each miRNA may have hundreds or even more potential targets. Therefore, for each additional miRNA to be further filtered out, it can save us a large amount of human labor and time.

C. Step 3: Further filtering miRNAs through semantics-driven analysis in OmniSearch.

OmniSearch is a miRNA-specific analytical software tool based upon domain ontologies and semantic technologies. For each miRNA returned from the Venn-diagram screening, we conducted comprehensive semantic searches and analysis in the OmniSearch tool. Besides obtaining a collection of targets (validated or putative) for each candidate miRNA, a rich set of additional data for each of the miRNA targets was also retrieved to provide us with extra clues in better understanding specific functions of the miRNA under consideration. Such additional information includes GO annotations, PubMed publications, relevant MeSH terms, RNAcentral sequence annotations, and involved pathways. Based on the comprehensive analysis of the federated knowledge returned in OmniSearch, we further filtered out most likely miRNAs as candidate biomarkers in GBM.

Note that during our prior work [22-24], both effectiveness and efficiency of the OmniSearch software were thoroughly evaluated by different groups of domain experts from numerous institutions and, within various fields.

Results and Discussion

A. DEM identification and analysis in GEO datasets.

We chose and downloaded three miRNA expression microarray series of GBM from GEO: (1) GSE32466 [27] is a miRNA expression profile between primary and recurrent GBM tissues from the same patients. (2) GSE51332 [28] is a miRNA expression profile of peripheral blood samples comparing healthy donors with GBM donors. (3) GSE62381 [29] is a set of differentially expressed CSF miRNAs in diagnosing central nervous system (CNS) malignancies, and we chose the data of CSF samples between the normal/non-disease group and the GBM group. Our selection of these three GEO series can well cover various aspects of GBM disease and its

different stages. Due to the space limit, we use the Series GSE32466 as an example to report our results as follows: Figure 1 exhibits details of the Series GSE32466 in the GEO repository. Table 1 shows the grouping information for GSE32466 – there are 12 samples for each of the two groups (Primary GBM vs. Recurrent GBM). Table 2 displays the list of DEMs from the Series GSE32466: a total of 250 DEMs sorted by significance (the first 19 are shown in the table). Note that the column “t” in Table 2 refers to moderated t-statistic [30][31] and the column “B” refers to B-statistic or log-odds [32][33] that the miRNA is differentially expressed. Figure 2 visualizes – using the GEO Profile graphics – different expression levels of hsa-miR-142-3p, which is one of the 250 DEMs obtained from GSE32466. In Figure 2, it is clearly demonstrated that expression of hsa-miR-142-3p was significantly different between primary (left 12 bars) and recurrent (right 12 bars) GBM tissues.

B. Initial screening through Venn diagrams.

Figure 3 exhibits the initial screening results via the application of Venn diagrams in Venny 2.1, including intersections between pairwise GEO series, as well as common intersections among all three GEO series. There were 33 common DEMs between GSE32466 and GSE51322, 55 common DEMs between GSE32466 and GSE62381, 34 common DEMs between GSE51322 and GSE62381; and finally, there were 10 common DEMs among all three GEO series: hsa-miR-142-3p, hsa-miR-124, hsa-miR-140-5p, hsa-miR-939, hsa-miR-663, hsa-miR-503, hsa-miR-1281, hsa-let-7c, hsa-miR-15a, and hsa-let-7. These ten miRNAs are likely regulating miRNAs in GBM, subject for further semantic analysis.

C. Further filtering: OmniSearch semantic analysis.

We conducted semantic searches and analysis in OmniSearch for each of the ten miRNAs, and filtered out three most likely regulating miRNAs: **hsa-miR-142-3p**, **hsa-let-7c**, and **hsa-miR-140-5p**, which we believed are strongly potential biomarkers in GBM patients. Note that six domain experts participated in conducting semantic searches and analysis using the OmniSearch tool. These experts are the first five co-authors plus the eighth co-author of this paper, who are clinical investigators in GBM and/or biomedical scientists in miRNA research.

Our further filtering was based on the following observations: (1) There was evidence for all three selected miRNAs to support their regulation on some target genes that were known to be biomarkers in some tumor. (2) Such miRNA::target regulation was either biologically validated, computationally predicted, or both. Validated regulation was demonstrated by related PubMed publications (Column 8 in Figure 4) and/or records from miRTarBase, a miRNA target validation database (Column 7 in Figure 4). Predicted regulation was retrieved from numerous miRNA target prediction databases (Columns 4–6 in Figure 4). (3) In addition, associated GO annotations and relevant pathway analysis further support that these three miRNAs are all closely related to some tumor. For example, Figure 5 shows that some target genes of hsa-miR-142-3p are involved in WNT signaling pathway that relates to embryonic development in the nervous system and is central to a number of brain pathologies; Figure 6 demonstrates GO annotations for some target genes of hsa-let-7c regarding oncogene-induced cell senescence; and Figure 7 shows that some target genes of hsa-miR-140-5p are involved in Notch signaling pathway whose up-regulation in cancer cells greatly affects chemoresistance. (4) The fact that these three miRNAs are closely related to some tumor suggests the higher likelihood that they

may be related to GBM as well. (5) On the other hand, there was no evidence in OmniSearch to support the direct association between any of these three miRNAs and GBM itself. Putting these observations together, it is reasonable to speculate that these three miRNAs could very likely be novel biomarkers not yet known in GBM and worthwhile for further investigation.

Conclusions

GBM bears a dismal prognosis with rapid relapse even after complete resection and radio chemotherapy. To find a set of biomarkers for early detection of GBM is still a primary goal to improve diagnosis and treatment of this deadly disease. On the other hand, miRNAs have been reported to serve as non-invasive biomarkers for diagnosis and prognosis in many tumor types other than GBM. This observation motivated us to develop effective computational methodologies to help identify likely serum miRNAs as candidate biomarkers in GBM. We first identified a collection of DEMs in GBM based on the dataset analysis in the GEO repository; we then used Venn diagrams to filter out more likely miRNAs out of originally discovered DEMs; and finally, we utilized OmniSearch, a semantic search and analytical software tool, to conduct semantics-oriented searches and analysis on the set of miRNAs filtered by Venn diagrams. Our comprehensive analysis results led to a promising conclusion in the present study: three miRNAs, hsa-miR-142-3p, hsa-let-7c, and hsa-miR-140-5p, were identified to be closely related to GBM, thus potentially serving as novel biomarkers in this important disease.

In the current study, Venn diagrams were utilized to analyze intersection among DEMs from multiple GEO data sets. Two alternatives are the UpSetR package [34][35] and Fisher's method (a.k.a. Fisher's combined probability test) [36][37]. In future work, we will explore these two methods and compared new results with those obtained in the current study. Besides, we also plan to conduct wet-lab validation on biological functions of the three miRNAs reported in this paper. We will perform luciferase reporter assays to confirm the suppression of target mRNAs by their paired miRNAs, followed by soft agar assays to assess the role of these miRNA::mRNA pairs in the malignant transformation of neuroepithelial cells.

List of abbreviations (optional section)

CNS	central nervous system
CSF	cerebrospinal fluid
CT	computer tomography
DEMs	differentially expressed miRNAs
EVs	extracellular vesicles
GBM	glioblastoma multiforme
GEO	Gene Expression Omnibus
GO	Gene Ontology
logFC	log fold change
MIAME	minimum information about a microarray experiment
miRNA/miR	microRNA
MRI	magnetic resonance imaging
NCBI	National Center for Biotechnology Information
NCI	The National Cancer Institute
NIH	The National Institutes of Health
PANTHER	Protein ANalysis THrough Evolutionary Relationships
WHO	World Health Organization

Declarations

Authors' contributions

HJ and ZY collected the dataset and drafted the manuscript together. HJ, ZY, XW, WC, SG, and GMB performed the data analysis and helped with making the tables and figures. JH conceived the study and revised the manuscript. MVK and SL participated in the manuscript preparation. All authors read and approved the final manuscript.

Authors' Information (optional section)

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Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

GSE32466, GSE51332, and GSE62381 are available from the GEO repository. The OmniSearch software tool is available at: <http://omnisearch.soc.southalabama.edu/ui/>.

Competing interests

There are no conflicts of interests.

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Figures

Figure 1 - Greater details of the Series GSE32466 in the GEO repository

Series GSE32466		Query DataSets for GSE32466
Status	Public on Sep 28, 2014	
Title	miRNA profiles in Primary and Recurrent GBM	
Organism	Homo sapiens	
Experiment type	Non-coding RNA profiling by array	
Summary	Glioblastoma (GBM) bears a dismal prognosis with rapid relapse following complete resection and radiochemotherapy. The involvement of microRNAs in tumor progression has been demonstrated in hepatoma, breast cancer, and prostate cancers. However, the microRNAs involved in modulating the progression and relapse of GBM are still unclear. Initially, we compared the miRNA expression profiles between primary and recurrent GBM tissues from the same patient in twelve independent cases.	
Overall design	miRNA expression profiles between primary and recurrent GBM tissues from the same patient in twelve independent cases.	
Contributor(s)	Chiou S	
Citation missing	Has this study been published? Please login to update or notify GEO.	
Submission date	Sep 29, 2011	
Last update date	Feb 06, 2015	
Contact name	Shih-Hwa Chiou	
E-mail	shchiou@vghtpe.gov.tw	
Organization name	Vghtpe	
Street address	No. 201, Sec. 2, Shih-Pai Road	
City	Taipei	
ZIP/Postal code	112	
Country	Taiwan	
Platforms (1)	GPL10850 Agilent-021827 Human miRNA Microarray (V3) (miRBase release 12.0 miRNA ID version)	
Samples (24)	GSM803543 GBM1 More... GSM803544 GBM2 GSM803545 GBM3	
Relations	PRJNA147879	

Figure 2 - Different expression levels of hsa-miR-142-3p (one of the 250 DEMs from GSE32466) using the GEO Profile graphics

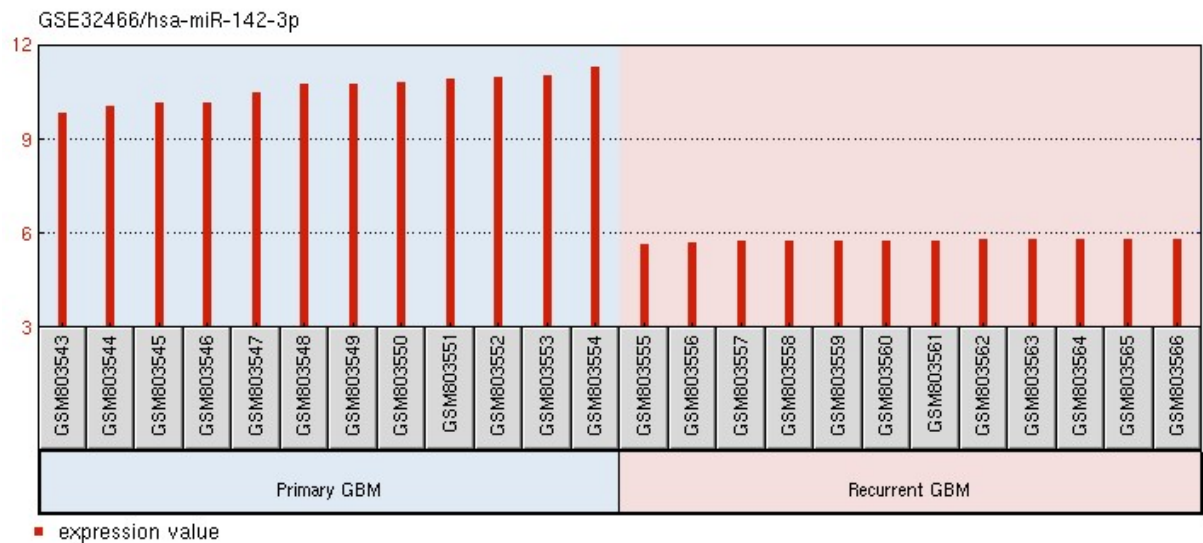


Figure 3 - Initial screening results via the application of Venn diagrams (using Venny 2.1): a total of ten regulating miRNAs in GBM

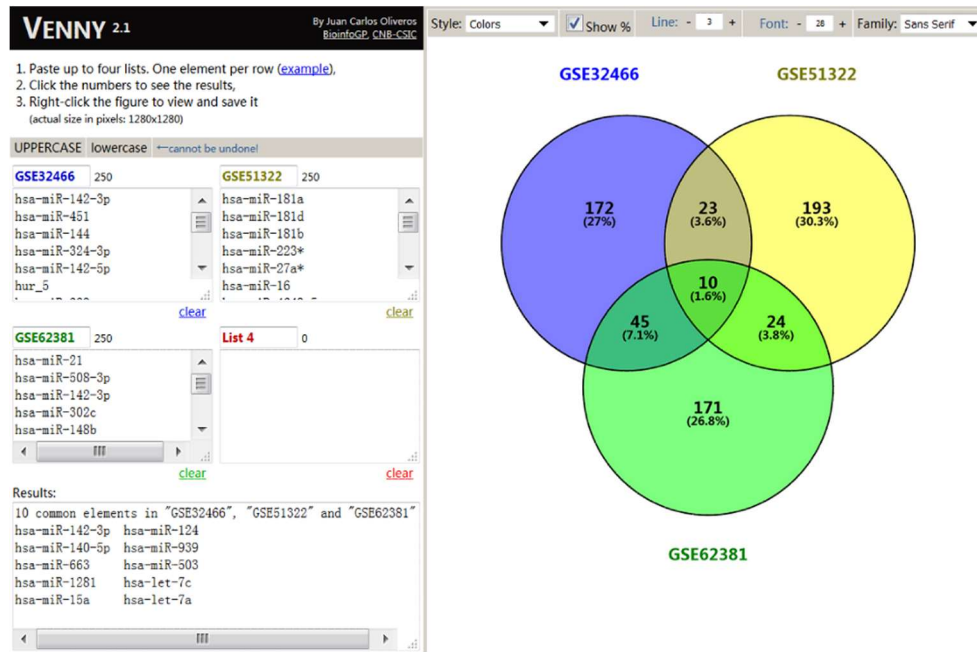


Figure 4 - Semantic search results for hsa-miR-142-3p with the MeSH term “Biomarkers, Tumor” in OmniSearch interface

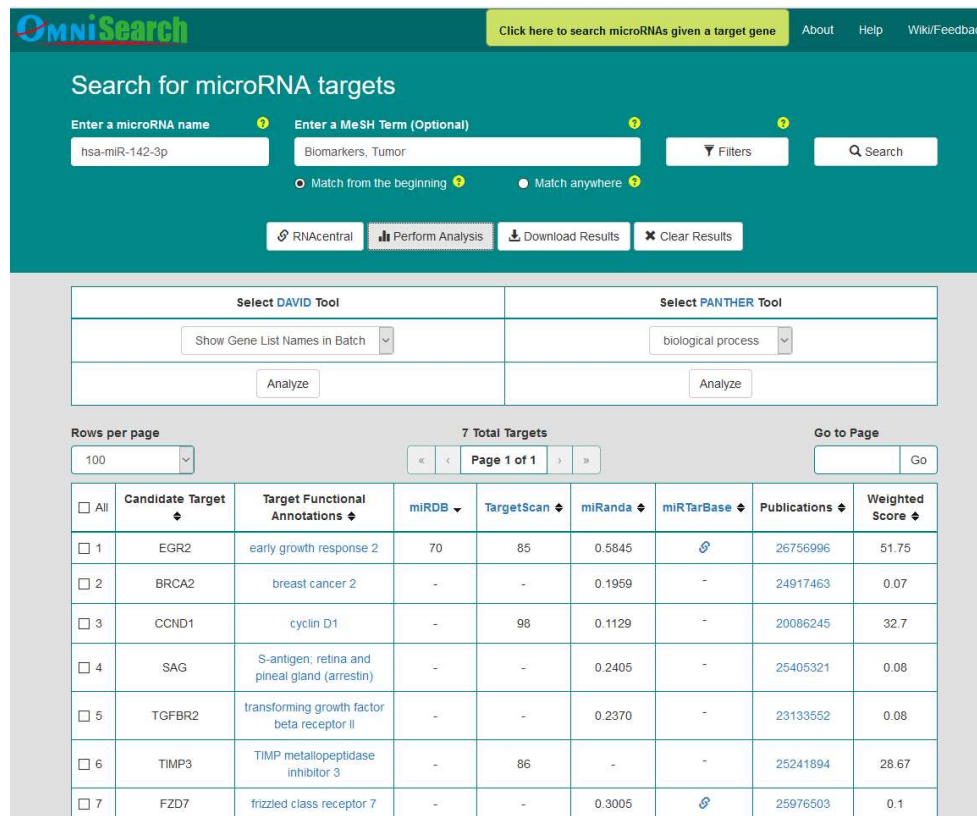


Figure 5 - PANTHER pathway analysis on some target genes of hsa-miR-142-3p: WNT signaling pathway

Displaying only results with $P < 0.05$; [click here to display all results](#)

	Homo sapiens (REF)	upload 1				
PANTHER Pathways	#	#	expected	Fold Enrichment	+/-	P value
Wnt signaling pathway	312	3	.12	25.24	+	2.78E-02
Unclassified	18373	4	7.00	.57	-	0.00E00

**Figure 6 - GO annotations for some target genes of hsa-let-7c
oncogene-induced cell senescence**

Term Information

Accession

GO:0090402

Name

oncogene-induced cell senescence

Ontology

biological_process

Synonyms

OIS

Alternate IDs

None

Definition

A cellular senescence process associated with the dismantling of a cell as a response to oncogenic stress, such as the activation of the Ras oncogenic family. Source: GOC:BHF

Comment

None

History

See term history for GO:0090402 at QuickGO

Subset

None

Related

[Link](#) to all **genes and gene products** annotated to oncogene-induced cell senescence.
[Link](#) to all direct and indirect **annotations** to oncogene-induced cell senescence.
[Link](#) to all direct and indirect **annotations download** (limited to first 10,000) for oncogene-induced cell senescence.

Annotations

Graph Views

Inferred Tree View

Neighborhood

Mappings

Filter results

Total annotations: 15; showing: 1-10

Results count [10](#)

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[<Prev](#)
[Next>](#)
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Figure 7 - PANTHER pathway analysis on some target genes of hsa-miR-140-5p: Notch signaling pathway

Displaying only results with $P < 0.05$; [click here to display all results](#)

	Homo sapiens (REF)	upload 1				
PANTHER Pathways	#	#	expected	Fold Enrichment	+/-	P value
Notch signaling pathway	44	2	.02	86.79	+	3.81E-02
Unclassified	18373	1	9.62	< 0.2	-	0.00E00

Tables

Table 1 - Grouping for the GEO series GSE32466: 12 samples for the primary and recurrent GBM groups, respectively

Group	Accession	Title	Source name	Cell type
Primary GBM	GSM803543	GBM1	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803544	GBM2	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803545	GBM3	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803546	GBM4	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803547	GBM5	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803548	GBM6	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803549	GBM7	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803550	GBM8	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803551	GBM9	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803552	GBM10	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803553	GBM11	Glioblastoma	Primary GBM tissue
Primary GBM	GSM803554	GBM12	Glioblastoma	Primary GBM tissue
Recurrent GBM	GSM803555	rGBM1	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803556	rGBM2	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803557	rGBM3	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803558	rGBM4	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803559	rGBM5	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803560	rGBM6	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803561	rGBM7	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803562	rGBM8	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803563	rGBM9	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803564	rGBM10	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803565	rGBM11	Recurrent Glioblastoma	Recurrent GBM tissue
Recurrent GBM	GSM803566	rGBM12	Recurrent Glioblastoma	Recurrent GBM tissue

Table 2 - The list of DEMs from the GEO series GSE32466: a total of 250 DEMs sorted by significance (the first 19 are shown in the table)

adj.P.Val: P-value after adjustment for multiple testing, recommended as the primary statistic to interpret results.

P.Value: Raw P-value.

t: Moderated t-statistic.

B: B-statistic or log-odds that the miRNA is differentially expressed.

logFC: Log2-fold change between two experimental conditions.

ID	adj.P.Val	P.Value	t	B	logFC	miRNA_ID	SPOT_ID
hsa-miR-142-3p	4.59e-19	4.77e-22	-37.26	40.079	-4.8289	hsa-miR-142-3p	
hsa-miR-451	1.31e-15	2.73e-18	-25.34	31.9378	-9.4375	hsa-miR-451	
hsa-miR-144	1.10e-14	3.45e-17	-22.59	29.4566	-4.2341	hsa-miR-144	
hsa-miR-324-3p	4.40e-14	1.83e-16	20.94	27.806	2.4653	hsa-miR-324-3p	
hsa-miR-142-5p	1.02e-11	5.31e-14	-16.09	22.1316	-1.3566	hsa-miR-142-5p	
hur_5	1.94e-10	1.21e-12	13.86	18.9686	2.6552		control
hsa-miR-223	2.14e-09	1.56e-11	-12.22	16.3688	-3.1405	hsa-miR-223	
mr_1	4.08e-09	3.40e-11	11.75	15.5759	3.7889		control
hsa-miR-765	4.54e-09	4.25e-11	-11.62	15.3475	-1.0929	hsa-miR-765	
hcmv-miR-UL70-3p	1.15e-08	1.22e-10	-11.01	14.2743	-1.588	hcmv-miR-UL70-3p	N/A
hsa-miR1274a	1.15e-08	1.32e-10	10.97	14.1939	3.9271	hsa-miR1274a	
hsa-miR-181c	4.33e-08	6.30e-10	-10.11	12.5981	-1.3197	hsa-miR-181c	
hsa-miR-146b-5p	4.82e-08	7.52e-10	-10.02	12.4183	-1.578	hsa-miR-146b-5p	
dmr_6	5.48e-08	9.59e-10	-9.89	12.1696	-4.8915		control
dmr_285	5.48e-08	9.70e-10	-9.88	12.1579	-1.3116		control
dmr_3	6.31e-08	1.18e-09	-9.78	11.9563	-4.5936		control
hsa-miR-1260	7.74e-08	1.53e-09	9.64	11.6927	2.5897	hsa-miR-1260	
SCorner3	8.44e-08	1.76e-09	-9.57	11.5516	-4.5773		control
hsa-miR-124	1.04e-07	2.49e-09	-9.39	11.1957	-4.2293	hsa-miR-124	

Additional files

None.